

Comment	Designator	Footprint
LED_BLUE	D37	LED_0402
LED_GREEN	D38	LED_0402
LED_RED	DS1, DS2, DS3, DS4, DS5, DS6, D	LED_0402
LED_YELLOW	D36	LED_0402
0R	R63, R323, R331, R333, R338	RES_C_1005X40
0R	R351, R352	RES_C_0603X30
0.01uF/16V	C7, C158	CAP_C_1608X80
0.1uF/10V	C15, C23, C24, C25, C26, C27, C2	CAP_C_1005X50_L
0.1uF/10V	C120, C183, C186, C346, C347, C	CAP_C_0603x30
0.1uF/10V	C151, C152, C232, C233, C234, C	CAP_C_0603x30
1k	R9, R138, R222, R314, R402, R42	RES_C_1005X40
1k	R335, R542, R543	RES_C_0603X30
1M 1%	R280, R324, R327	RES_C_1005x30_L
1N4007FL	D5, D6, D7, D8, D9, D10, D11, D1	SOD3717X145N
1uF/20V	C4, C9, C13, C61, C62, C63, C64,	CAP_C_1005X50_L
1uH	L2, L3, L4, L5	L-SMD-M-3.5x3.5x2mm
1.6k	R6, R209, R255, R341, R423, R43	RES_C_1005X40
1.69k	R557	RES_C_1005X40
2.2nF/10V	C5, C6, C145, C146, C166, C167,	CAP_C_1005X50_L
3.24k	R8, R10, R253, R256, R298, R413	RES_C_1005X40
4.02k	R321	RES_C_0603X30
4.7k	R141, R285, R290	RES_C_0603X30
4.7k	R192, R197, R198, R199, R200, R	RES_C_1005X40
4.7uF/10V	C41, C47, C48, C51, C52, C69, C7	CAP_C_1005X50_L
5.62k	R110	RES_C_1005X40
10k	R4, R7, R61, R111, R112, R113, R	RES_C_1005X40
10k	R302, R340, R342, R346, R348, R	RES_C_0603X30
10M	R109	RES_C_1005X40
10uF/20V	C10, C21, C22, C147, C343	CAP_C_2012X110
12pF/10V	C18, C19, C20	CAP_C_1005X50_L
16.5k/1%	R409, R410	RES_C_1005X40
16.9k	R311, R317	RES_C_0603X30
18pF/10V	C16, C17, C265, C272, C276, C28	CAP_C_0603x30
20k	R313, R315	RES_C_0603X30
20R/2W/5V or 10R/1W/3V	R90, R92, R284, R334, R488, R50	RES_C_6331X65_N
22R	R11, R12, R13, R14, R15, R16, R1	RES_C_1005X40
22R	R62, R168, R169, R170, R171, R1	RES_C_0603X30
22uF/10V	C282, C288, C290, C292	CAP_C_1608X80
47k	R2, R3, R5, R136, R137, R212, R2	RES_C_1005X40
47nF/10V	C43, C44, C45, C46, C55, C56, C6	CAP_C_1005X50_L
47uF/25V	C1, C2, C3, C8, C11, C12, C14, C	CAP_C_3225X80
60R	R441, R442	RES_C_1005X40
100k	R339, R540	RES_C_0603X30
100R	R114, R119	RES_C_1005X40
120R	R152, R153, R154, R158, R159, R	RES_C_1005X40
133k	R204, R224, R254, R422, R430, R	RES_C_1005X40
150pF/10V	C213, C215, C217, C218, C219, C	CAP_C_1005X50_L

Comment	Designator	Footprint
200R	R301, R307, R310	RES_C_0603X30
240R	R539, R541	RES_C_0603X30
470k	R1, R129, R211, R249, R417, R42	RES_C_1005X40
0473460001	J16	47346-0001_MOL
0532610271	J26	molex_0532610271
0532610471	J15, J18, J19	MOLEX_0532610471
0532610571	J4, J5	MOLEX_0532610571
0532610671	J13, J14, J27, J28	MOLEX_0532610671
0532610971	J21, J22	MOLEX_0532610971
0532611171	J23, J24	MOLEX_0532611171
1051620001	J17	Molex_1051620001
5040500291	J31	MOLEX_5040500291
5040500691	J34	MOLEX_5040500691
5040500891	J2, J6, J7, J8, J9, J10, J11, J12, J	MOLEX_5040500891
5040500991	J1	MOLEX_5040500991
AO3400A	Q3, Q4, Q5, Q6, Q7, Q8, Q9, Q10	SOT95P240X120-3N
AS3004204-0108X0PSAY	U30, U42	SOIC127P790X216-8
BMP390	U48	10LGA_2X2X0p75_BOS
BSS138NH6327XTSA2	Q27	SOT-23-3
Crystal 2 Pins	Y1, Y5	CRYSTAL_NX2012SA
Crystal 4 Pins	Y3	CRYSTAL_ABM8G
DFE252012P-1R5M=P2	L7, L8, L9, L10	IND_C_1008
DMG302PU-7	Q1, Q2	SOT95P240X120-3N
DNP/0R	JP1, JP2, JP3, JP4, JP5, JP6, JP7	JUMPER_1608
DNP/0R	R120, R121, R332	RES_C_1005X40
DNP/0R	R325, R326, R328	RES_C_0603X30
DNP/4.7k	R193, R194, R195, R196, R289, R	RES_C_1005X40
DNP/10k	R128, R210, R227, R228, R229, R	RES_C_1005X40
DNP/10k	R303, R306, R390, R391, R460, R	RES_C_0603X30
DNP/22R	R83	RES_C_1005X40
DSC10xx DFN-4	Y2	DFN-4_2.5x2
DSC60xx DFN-4_LGA-4	Y4	DFN-4_2x1.6
FC5BQCCMC12.0-T1	XC2	AP7-00039
FSUSB74MPX	U4	QFN50P300X300X80-17N
Header 8x1	J3, J25	Header_254P_8x1
IP4220CZ6	U21, U22	SOT95P280X145-6N
IS43TR16512BL-125KBLI	U53	TWBGA96_ISI
IS66WVS8M8FBLL-104NL	U33	SOIC8_150MIL_N_ISI
KDZLVTFTTR51	DZ1, DZ2, DZ3, DZ4, DZ5, DZ6, D	SOD3717X145N
KRL1632E-M-R010-F-T5	R344, R349, R353, R354	RES_C_1206_SHUNT
LM1117MP-3.3/NOPB	VR1, VR2	VREG_LM1117MP-3.3/NOPB
LSM6DSOXTR	U49	PQFN50P250X300X86-14N
LT8722AV#PBF	U6, U7, U8, U9	QFN18
LTC2865HMSE#TRPBF	U54	MSOP-12_MSE_LIT
M3	H3, H4, H7, H8	HEX-M3_NoVia
MAX3488ESA+T	U10, U57	SOIC127P600X175-8N
MAX14662ETI+T	U56	QFN40P400X400X80-29N
MCP16502 QFN-32	U51	QFN-32_5X5_EP3.7X3.7

Comment	Designator	Footprint
MIC2026-1YM	U5	SOIC127P602X173-8N
MPD-200A	U50	MPD-200a
mp-HighDriver4	U16	MP_HIGHDRIVER4
MTFC32GAZAQDW-AAT	U23	BGA_100_C_100P_10X17_1400X18
OPA2340UA	U11, U12, U13, U14	SOIC127P602X173-8N
PAC1934 UQFN-16	U52	UQFN-16_4X4_EP2.6X2.6
PCA9508DP,118	U17, U18, U19, U20, U45, U46	TSOP8
RV-3129-C3-32.768KHZ-O	U31	IC_RV-3129-C3_32.768KHZ_OPTIO
SAMA7G54	U29	TFBGA-343_14x14
SAMV71Q21B	U3	AP6-00118
SN65HVD230DR	U39, U40, U41	SOIC127P599X175-8N
SN3257QPWRQ1	U32	SOIC_PRQ1_TEX
SRP7050TA-220M	L1, L6	PSM6050
SS54	D1, D2, D13, D14, D15, D16, D17,	DO-214AC_SMA
TB67H451AFNG,EL	U25, U26, U27, U28	SOIC127P602X173-8N_EPAD
TC1047AVNBTR	U15	SOT95P240X120-3N
TCA6416APWR	U47	SOP65P440W-24N
TPL5010QDDCTQ1	U34	SOT95P280X145-6N
TPS5430DDAR	U2, U35	TPS5430DDAR_V3
TPS74901RGWR	U55	QFN65P500X500X100-21N
TPS259470ARPWR	U1, U24, U36, U37, U38, U43, U44	VQFN-HR-10
USBLC6-2SC6	D3, D4	SOT95P280X145-6N

Description	Quantity	TargetPr	DesignItemId	SourceLibraryName	Manufacturer 1
LED	1	0	LED_BLUE	LABLIB.SchLib	Lite-On
LED	1	0	LED_GREEN	LABLIB.SchLib	Lite-On
LED	10	0	LED_RED	LABLIB.SchLib	Lite-On
LED	1	0	LED_YELLOW	LABLIB.SchLib	Lite-On
RES	1	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	6	0	RES_B_2	LABLIB.SchLib	Panasonic
CAP	2	0	CAP_C_2	LABLIB.SchLib	Samsung Electro-M
CAP	108	0	CAP_C_2	LABLIB.SchLib	Samsung Electro-M
CAP	38	0	CAP_C_2	LABLIB.SchLib	Murata
CAP	50	0	CAP_C_2	LABLIB.SchLib	Murata
RES	5	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	3	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	3	0	RES_B_2	LABLIB.SchLib	Panasonic
DIODE	24	0	Diode SMD	LABLIB.SchLib	SMC Diode Solution
CAP	29	0	CAP_C_2	LABLIB.SchLib	Holy Stone Enterpris
IND	4	0	L-SMD-1	EXP_HW_V3.SchLib	Bourns
RES	7	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	1	0	RES_B_2	LABLIB.SchLib	Stackpole Electronic
CAP	14	0	CAP_C_2	LABLIB.SchLib	Samsung Electro-M
RES	11	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	1	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	3	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	61	0	RES_B_2	LABLIB.SchLib	Panasonic
CAP	29	0	CAP_C_2	LABLIB.SchLib	Samsung Electro-M
RES	1	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	113	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	11	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	1	0	RES_B_2	LABLIB.SchLib	Panasonic
CAP High Value	5	0	CAP_C_High_2	LABLIB.SchLib	Samsung Electro-M
CAP	3	0	CAP_C_2	LABLIB.SchLib	Murata
RES	2	0	RES_B_2	LABLIB.SchLib	Stackpole Electronic
RES	2	0	RES_B_2	LABLIB.SchLib	Panasonic
CAP	6	0	CAP_C_2	LABLIB.SchLib	KEMET
RES	2	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	16	0	RES_B_2	LABLIB.SchLib	Stackpole Electronic
					Bourns Inc.
RES	133	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	82	0	RES_B_2	LABLIB.SchLib	Yageo Group
CAP	4	0	CAP_C_2	LABLIB.SchLib	Murata
RES	15	0	RES_B_2	LABLIB.SchLib	Panasonic
CAP	12	0	CAP_C_2	LABLIB.SchLib	Samsung Electro-M
CAP High Value	29	0	CAP_C_High_2	LABLIB.SchLib	TAIYO YUDEN
RES	6	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	2	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	2	0	RES_B_2	LABLIB.SchLib	Vishay
RES	10	0	RES_B_2	LABLIB.SchLib	Panasonic
RES	6	0	RES_B_2	LABLIB.SchLib	Panasonic
CAP	16	0	CAP_C_2	LABLIB.SchLib	Samsung Electro-M

Description	Quantity	TargetPr	DesignItemId	SourceLibraryName	Manufacturer 1
RES	3	0	_RES_B_2	LABLIB.SchLib	Yageo Group
RES	2	0	_RES_B_2	LABLIB.SchLib	Stackpole Electronic
RES	7	0	_RES_B_2	LABLIB.SchLib	Panasonic
No Description	1	0	USB_micro	obc_ver2.SCHLIB	Molex
	1	0	Header 2x1_2M	LABLIB.SchLib	Molex
	3	0	Header 4x1_2M	LABLIB.SchLib	Molex
	2	0	Header 5x1_2M	LABLIB.SchLib	Molex
	4	0	FLOW_SENSOR	LABLIB.SchLib	Molex
	2	0	Header 9x1_2M	LABLIB.SchLib	Molex
	2	0	Header 11x1_2M	LABLIB.SchLib	Molex
9 (8 + 1) Posit	1	0	Molex_1051620001	new_component.SchLib	Molex
	1	0	Header 2x1_2M	LABLIB.SchLib	Molex
	1	0	Header 6x1_2M	LABLIB.SchLib	Molex
	13	0	Header 8x1_2M	LABLIB.SchLib	Molex
	1	0	Header 9x1_2M	LABLIB.SchLib	Molex
N-Channel 30	24	0	MOSFET_N	LABLIB.SchLib	Alpha & Omega Ser
	2	0	IC_RAM_AS3004204-0	LABLIB.SchLib	Avalanche Technolo
Description	1	0	BMP390	LABLIB.SchLib	Bosch Tools
Transistor MO	1	0	TRA1223	LABLIB.SchLib	Infineon
Crystal	2	0	OSC_32.768kHz	LABLIB.SchLib	NDK
Crystal	1	0	OSC_24MHz	LABLIB.SchLib	Abrakon
Inductor	4	0	IND1335	LABLIB.SchLib	Murata
	2	0	MOSFET_P_GSD	LABLIB.SchLib	Diodes Inc.
	11	0	JUMPER	EXP_V1.SchLib	
RES	2	0	_RES_B_2	LABLIB.SchLib	Panasonic
RES	4	0	_RES_B_2	LABLIB.SchLib	
RES	6	0	_RES_B_2	LABLIB.SchLib	Panasonic
RES	17	0	_RES_B_2	LABLIB.SchLib	Panasonic
RES	11	0	_RES_B_2	LABLIB.SchLib	Panasonic
RES	1	0	_RES_B_2	LABLIB.SchLib	Panasonic
	1	0	MIC6659	LABLIB.SchLib	Microchip Technolog
	1	0	MIC8888	LABLIB.SchLib	Microchip
Fox FQ5032B	1	0	A07-0042	AltiumDbLib.DbLib	Abrakon
USB Switch IC	1	0	IC_USB_SW_FSUSB7	LABLIB.SchLib	onsemi
	2	0	Header 8x1	LABLIB.SchLib	Samtec Inc.
	2	0	IP4220CZ6_0	LABLIB.SchLib	Nexperia
DDR3L	1	0	IC_RAM_IS43TR16512	LABLIB.SchLib	ISSI
No Description	1	0	IS66WVS4M8BLL-104M	LABLIB.SchLib	ISSI
Zenner	16	0	DIODE Zener	LABLIB.SchLib	ROHM
Shunt	4	0	RSMT1316	LABLIB.SchLib	Susumu
800-mA 15-V	2	0	LM1117MP-3.3/NOPB	LABLIB.SchLib	Texas Instruments
Accelerometer	1	0	LSM6DSOXTR	LABLIB.SchLib	STMicroelectronics
	4	0	LT8722	EXP_HW_V3.SchLib	Analog Devices
±60V Fault P	1	0	LTC2865HMSETRPBF	LABLIB.SchLib	Analog Devices Inc.
	4	0	HEX-M3	hex_M3.IntLib	
1/1 Transceiver	1	0	MAX3488ESA+T	LABLIB.SchLib	Analog Devices
8 Circuit IC Sw	1	0	IC_SW_MAX14662ETI	LABLIB.SchLib	Analog Devices
	1	0	MIC8878	LABLIB.SchLib	Microchip Technolog

Description	Quantity	TargetPr	DesignItemId	SourceLibraryName	Manufacturer 1
Power Switch/	1	0	IC_PWRSW_MIC2026	LABLIB.SchLib	Microchip
	1	0	MPD-200A	LABLIB.SchLib	
	1	0	mp_HighDriver4	new_component.SchLib	
	1	0	MTFC32GAZAQDW-A	emmc_nofootprint.IntLib	Micron
	4	0	OPA2340UA	EXP_HW_V3.SchLib	Texas Instruments
	1	0	MIC8883	LABLIB.SchLib	Microchip
	6	0	PCA9508DP,118	EXP_HW_V3.SchLib	NXP Semiconducto
Real-Time Clo	1	0	RV-3129-C3_32.768KH	RV-3129-C3_32.768KH	Micro Crystal AG
SAMA7G54 M	1	0	SAMA7G54	LABLIB.SchLib	Microchip
SAM 32-bit AF	1	0	A06-1541	AltiumDbLib.DbLib	Microchip
3.3 V CAN Tra	3	0	SN65HVD230DR	COMM_SCHLIB.SchLib	Texas Instruments
Automotive 5-	1	0	IC_SW_SPI_SN3257Q	LABLIB.SchLib	Texas Instruments
FIXED IND 22	2	0	SRP7050TA-220M	LABLIB.SchLib	Bourns
DIODE SCHO	9	0	DIODE SCHOTTKY	LABLIB.SchLib	EVVO
BRUSHED MO	4	0	IC_H_bridge_TB67H45	LABLIB.SchLib	Toshiba
Temperature S	1	0	IC_SENSOR_TC1047A	LABLIB.SchLib	Microchip
	1	0	IC_I2Cto16IO_TCA641	CAMERA_SchLib.SchLib	Texas Instruments
	1	0	TPL5010QDDCTQ1	TPL5010QDDCTQ1.IntLib	Texas Instruments
IC REG BUCK	2	0	TPS5430DDAR	EXP_HW_V3.SchLib	Texas Instruments
3-A, low-VIN (	1	0	IC_PWR_TPS74901RC	LABLIB.SchLib	Texas Instruments
	7	0	TPS259470ARPWR	EXP_HW_V3.SchLib	Texas Instruments
17V Clamp 5A	2	0	USBL6-2SC6	USBL6-2SC6.IntLib	STMicroelectronics

Manufacturer Part Number 1	Note
LTST-C281TBKT	
LTST-C281KGKT	
LTST-C281KRKT	
LTST-C281KSKT	
ERJ-3GEY0R00V	
ERJ-3GEY0R00V	
CL10B103KB8NNNC	
CL05B104KA5NNNC	
GRM033C71C104KE14D	
GRM033C71C104KE14D	
ERJ-2RKF1001X	
ERJ-1GNF1001C	
ERJ-2RKF1004X	
1N4007FL	
C0402B105K025T	
SRP3020TA-1R0M	
ERJ-2RKF1601X	
RMCF0402FT1K69	
CL05B222KB5NNNC	
ERJ-2RKF3241X	
ERJ-1GNF4021C	
ERJ-1GNF4701C	
ERJ-2RKF4701X	
CL05A475MP5NRNC	
ERJ-2RKF5621X	
ERJ-2RKF1002X	
ERJ-1GNJ103C	
ERJ-2GEJ105X	
CL21A106KAYNNNE	
GJM1555C1H120FB01D	
RMCF0402FT16K5	
ERJ1GNF1692C	
CBR04C180F5GAC	
ERJ-1GNJ203C	
RPC2512JT10R0	
CRM2512-FX-20R0ELF	
ERA-2AKD220X	
RC0201FR-0722RL	
GRM188R61A226ME15D	
ERJ-2RKF4702X	
CL05B473KA5VPNC	
TMK325ABJ476MM-P	
ERJ-2RKF60R4X	
ERJ-1GNF1003C	
CRCW0402100RFKEDHP	
ERA-2AEB121X	
ERJ-2RKF1333X	
CL05C151JB5NNNC	

Manufacturer Part Number 1	Note
AC0201FR-07200RL	
RMCF0201FT240R	
ERJ-2RKF4703X	
47346-0001	
53261-0271	
53261-0471	
53261-0571	
53261-0671	
53261-0971	
53261-1171	
105162-0001	
504050-0291	
504050-0691	
504050-0891	
504050-0991	
AO3400A	
AS3004204-0108X0PSAY	
BMP390	
BSS138NH6327XTSA2	
NX2012SA-32.768K-STD-MUB-1	
ABM8G-24.000MHZ-18-D2Y-T	
DFE252012P-1R5M=P2	
DMG302PU-7	
ERJ-3GEY0R00V	
ERJ-2RKF4701X	
ERJ-2RKF1002X	
ERJ-1GNJ103C	
ERA-2AKD220X	
DSC1001DI5-024.0000T	
DSC6003MA3B-032K768	
FC5BQCCMC12.0-T1	
FSUSB74MPX	
TSM-108-01-L-SV-P-TR	
IP4220CZ6F	
IS43TR16512BL-125KBLI	
IS66WVS8M8FBLL-104NLI	
KDZLVTFT51	
KRL1632E-M-R010-F-T5	
LM1117MP-3.3/NOPB	
LSM6DSOXTR	
LT8722AV#PBF	
LTC2865HMSE#TRPBF	
MAX3488ESA+T	
MAX14662ETI+T	
MCP16502TAB-E/S8B	

Manufacturer Part Number 1	Note
MIC2026-1YM-TR	
MTFC32GAZAQDW-AAT	
OPA2340UA	
PAC1934T-I/JQ	
PCA9508DP,118	
RV-3129-C3-32.768KHZ-OPTION-A-TB-QA	
SAMA7G54-V/4HB	
ATSAMV71Q21B-CB	
SN65HVD230DR	
SN3257QPWRQ1	
SRP7050TA-220M	
SS54	
TB67H451AFNG(O,EL)	
TC1047AVNBTR	
TCA6416APWR	
TPL5010QDDCTQ1	
TPS5430DDAR	
TPS74901RGWR	
TPS259470ARPWR	
USBLC6-2SC6	